

UNIT 1: Object Oriented Programming in C++

1. Object Oriented Programming (OOP)

Definition

Object Oriented Programming is a programming approach in which a program is designed using objects and classes. It focuses on data security, reusability and modularity.

Features of OOP

- Data Hiding
 - Encapsulation
 - Abstraction
 - Inheritance
 - Polymorphism
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2. Class and Object

Class

A class is a user-defined data type which contains data members and member functions.

Object

An object is an instance of a class. It represents a real-world entity.

Example

- Class → Student
 - Object → Afnur, Rahim
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3. Data Hiding

Definition

Data hiding means protecting data from outside access by using access specifiers like `private`.

Advantages

- Increases security
 - Prevents misuse of data
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4. Data Encapsulation

Definition

Encapsulation is the process of wrapping data and functions together into a single unit (class).

Example

Data members and member functions inside a class.

5. Abstraction

Definition

Abstraction means showing only essential details and hiding unnecessary details.

Example

Using an ATM machine without knowing its internal working.

6. Abstract Class and Concrete Class

Abstract Class

A class that cannot be instantiated (object cannot be created).

Concrete Class

A class from which objects can be created.

7. Polymorphism

Definition

Polymorphism means one function name performing different tasks.

Type (as per syllabus)

- Function Overloading

Function Overloading

When multiple functions have the same name but different parameters.

8. Inheritance

Definition

Inheritance is the process by which a derived class acquires the properties of a base class.

Advantages

- Code reusability
- Saves time
- Reduces redundancy

Types of Inheritance

- Single Inheritance
- Multilevel Inheritance
- Multiple Inheritance

9. Visibility Modes (Access Specifiers)

Specifier	Accessibility
private	Inside class only
public	Anywhere
protected	Inside class and derived class

Note: Default visibility mode in C++ class is **private**.

10. Member Functions

Member functions can be defined:

- Inside the class
- Outside the class using scope resolution operator `::`

11. Passing Objects to Functions

- **Pass by Value:** Copy of object is passed
 - **Pass by Reference:** Original object is passed
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12. Constructor

Definition

A constructor is a special member function which is automatically called when an object is created.

Characteristics

- Same name as class
- No return type
- Automatically invoked

Types of Constructor

- Default constructor
 - Parameterized constructor
 - Copy constructor
 - Constructor with default arguments
 - Overloaded constructor
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13. Destructor

Definition

A destructor is a special member function which is called automatically when an object is destroyed.

Characteristics

- Name starts with ~
 - No arguments
 - Only one destructor allowed
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14. File Handling

Need of File

- Permanent storage of data
- Easy retrieval

Types of Files

- Text File
 - Binary File
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15. Text File

Operations

- Create
- Write
- Read
- Modify

Functions Used

- `open()`
 - `close()`
 - `get()`
 - `put()`
 - `getline()`
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16. Binary File

Operations

- Write
- Read
- Search
- Append
- Insert
- Delete
- Modify

Functions

- `read()`
 - `write()`
 - `seekg()`
 - `seekp()`
 - `tellg()`
 - `tellp()`
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17. Header Files for File Handling

- `ifstream` → Reading
 - `ofstream` → Writing
 - `fstream` → Both reading and writing
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18. Pointer

Definition

A pointer is a variable that stores the address of another variable.

Uses

- Dynamic memory allocation
- Efficient memory usage

Operators

- `*` → Dereference operator
 - `->` → Access structure members
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19. Dynamic Memory Allocation

- `new` → Allocate memory
 - `delete` → Deallocate memory
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Very Important Exam Points

- Define OOP, Class, Object (1–2 marks)
- Constructor vs Destructor (3–5 marks)
- Inheritance types and advantages
- File handling operations
- Pointer definition and uses